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Alinani Simukanga

Lecturer II

Academic Profile

Computer Science academic with teaching, research, and academic leadership experience across Software Engineering, Computer Systems, and Cybersecurity-related courses.

Research interests include ICT4D, biometric identification systems, and ethical AI in digital health contexts.

Active contributor to curriculum delivery, student supervision, and departmental quality assurance processes.

Current focus: strengthening practice-oriented teaching, industry engagement, and applied research impact.

Career

University of Zambia, Lusaka, Zambia | Lecturer II 15/12/2025 - Present

- Served in Acting Head of Department assignments (interim periods) with responsibility for day-to-day departmental academic coordination.
- Teaching *CSC 2101 - Computer Systems*, *CSC 3600 - Software Engineering*, *CSC 5515 - Emerging Technologies*, and *CSC 5801 - Programming for Data Science*.
- Supervising final year student projects *CSC 4004 - Project*.
- Assisting in the coordination of student internships *CSC 3009 - Industrial Training*.

University of Edenberg, Lusaka, Zambia | Lecturer B 08/07/2024 - 12/12/2025

- Taught *CSE111 - Introduction to Information Technology*, *CSE113 - Computer Architecture and Organization*, *CSE114 - Programming with Python*, *CSE122 - Object-Oriented Programming with C++*, *CSE124 - Operating Systems* and *CSE125 - Data Structures and Algorithms*.
- Held the role of Program Coordinator for Undergraduate and Postgraduate programs under the School of Computer Science & IT.

ZCAS University - Lusaka, Zambia | Part-time Lecturer 28/02/2024 - 03/01/2025

- Taught *Cyber Security Operations (CIT4351)* to full-time, part-time, and distance students (21). Organised and conducted laboratory sessions to familiarise students with commonly used tools in cybersecurity.
- Taught *Component and Connection Security (CIT3362)* and *IoT Fundamentals: IoT Security (CIT3352)* to full-time, part-time, and distance students (18).

EQUIP - Lusaka, Zambia | Application Developer 2020

- Assisted in developing the Project SEARCH application on the Android platform.
- Worked closely with personnel in Boston to analyse data, interpret analysis, and incorporate changes into the app.
- Worked with Dimagi to ensure smooth integration between the CommCare and SEARCH applications.

Project SEARCH - BU & UNZA | Researcher/Application Developer 2018 - 2023

- Worked with a multi-disciplinary team from Boston University and the University of Zambia to develop and refine a biometric tool targeting infants.
- Conducted tests on the longitudinal ear dataset to demonstrate application performance with growth as a factor.
- Formulated experiments to benchmark the performance of the matching algorithm.
- Authored a paper on the growth study results.

Career (Continued)

University of Zambia | Part-time Tutor 2017

- Provided support during the laboratory sessions of the Electronics course.
- Assisted in the supervision of a few undergraduate capstone projects.

Academic Leadership

University of Zambia, Lusaka, Zambia | Acting Head of Department (Interim) 2026 (Multiple Appointed Periods)

- Led departmental academic and administrative coordination, including teaching timetables and workload balancing.
- Supported curriculum quality assurance, moderation planning, and implementation of agreed academic action items.
- Chaired or coordinated departmental meetings and formal communication with faculty leadership and staff.
- Oversaw student support workflows and continuity of service delivery during interim leadership periods.

Teaching & Supervision

Core Teaching Areas: Software Engineering, Computer Systems, Programming, Cybersecurity, Data Structures, and Operating Systems.

Current Modules: *CSC 2101 - Computer Systems*, *CSC 3600 - Software Engineering*, *CSC 5515 - Emerging Technologies*, *CSC 5801 - Programming for Data Science*, and supervision in *CSC 4004 - Project*.

Project Supervision: Supervises undergraduate capstone and final-year projects, with focus on software quality, applied systems design, and industry relevance.

Academic Mentorship: Supports internship readiness, project planning, and professional development for undergraduate students.

Student Supervision

- **Undergraduate Final-Year Supervision**: Supervised 49 undergraduate final-year projects (selected examples below).
- **[Rishita Lalitbhai Thawani, Joylet Nduulwa, Faiz Padhar] AgroRent** – BSc Computer Science, University of Zambia, Supervisor, 2026 (Ongoing).
- **[Chilenje Victor] Smart Agriculture App - Mobile App that uses IoT sensors aand AI to help farmers monitor soil, crops, and weather** – BSc Information Technology, University of Lusaka, Supervisor, 2026 (Ongoing).
- **[Grace Musanga, Blessings M. Libonda, Mwangilwa Chishiba, Idah Mumbi and Sereni Banda] Inventory and Sales management System** – BSc Software Engineering, University of Zambia, Supervisor, 2026 (Ongoing).
- **[Chileshe Nduba] Insurance Choice Analyzer - An Intelligent System that compares insurance companies and policies, guiding customers toward the best option based on their needs** – BSc Information Technology, University of Lusaka, Supervisor, 2026 (Ongoing).
- **[Smart Mbuzi and Christopher Banda] Kajob** – BSc Software Engineering, University of Zambia, Supervisor, 2026 (Ongoing).
- **[Chitingiza Mapalo] AI-Assistend Career Path Advisor for Students** – BSc Information Technology, University of Lusaka, Supervisor, 2026 (Ongoing).
- Full supervision record available on request.

Education

Department of Computing and Informatics, School of Natural and Applied Sciences,
University of Zambia Lusaka, Zambia

Master of Science in Computer Science 2019

- Supervisor: Prof. Jackson Phiri & Dr. Mayumbo Nyirenda
- Topic: "Automation of the Grain Purchasing Process for Zambia's Food Reserve Agency"
- Research area: ICT4D and Remote Sensing

Department of Computing and Informatics, School of Natural and Applied Sciences,
University of Zambia Lusaka, Zambia

Bachelor of Science (Software Engineering) 2015

- Classification: Merit

Publications

1. Wenda Qin, Lauren Etter, **Alinani Simukanga**, Christopher Gill, & Margrit Betke. Age-constrained Ear Recognition: The EICZA Dataset and SASE Baseline Model. *2023 IEEE International Joint Conference on Biometrics (IJCB)*, 2023. https://doi.org/10.1109/IJCB57857.2023.10448594

2. Francis Kawesha, Jackson Phiri, & **Alinani Simukanga**. A Predictive Model to Support Decision Making for the Accreditation of Learning Programmes using Data Mining and Machine Learning. *Pan African Conference on Science, Computing and Telecommunications (PACT) 2023*, 2023. https://ictjournal.icict.org.zm/index.php/icict/article/view/286

3. **Alinani Simukanga**, Lauren Etter, Wenda Qin, Rachel Pieciak, Margrit Betke, William Macleod, Jackson Phiri & Christopher Gill. Ethical issues associated with the development of an ear biometric tool for patient identification in Zambia. *Global Forum for Bioethics in Research (GFBR)*, 2022. https://www.gfbr.global/wp-content/uploads/2022/12/Alinani-Simukanga_GFBR-2022_Theme-2.pdf

4. **Alinani Simukanga**, Misaki Kobayashi, Lauren Etter, Wenda Qin, Rachel Pieciak, Duarte Albuquerque, Yu-Jen Chen, Margrit Betke, William Macleod, Jackson Phiri, Lawrence Mwananyanda & Christopher Gill. The impact of ear growth on identification rates using an ear biometric system in young infants. *Gates Open Research*, 2021. https://doi.org/10.12688/gatesopenres.13459.1

5. Lauren Etter, **Alinani Simukanga**, Wenda Qin, Rachel Pieciak, Lawrence Mwananyanda, Margrit Betke, Jackson Phiri, Caroline Carbo, Arnold Hamapa & Christopher Gill. Project SEARCH (Scanning EARs for Child Health): validating an ear biometric tool for patient identification in Zambia. *Gates Open Research*, 2020. https://doi.org/10.12688/gatesopenres.13197.1

6. **Alinani Simukanga**, Jackson Phiri, Mayumbo Nyirenda, & Monica Kalumbilo-Kabemba. ICT in Governance Systems: A Case Study of the FISP Farmer Registration System. *ZAPUC*, 2018. https://scispace.com/pdf/e-governance-systems-a-case-study-of-the-development-of-a-1k42n7edea.pdf

7. Sipiwe Chihana, Jackson Phiri, Douglas Kunda, & **Alinani Simukanga**. Securing Grain in Transit for the Food Reserve Agency Based on the Cloud Model. *ZAPUC*, 2018.

8. **Alinani Simukanga**, Jackson Phiri, Mayumbo Nyirenda, & Monica Kalumbilo-Kabemba. E-Governance Systems: A Case Study of the Development of a Small-Scale Farmer Database. *Zambia Information Communication Technology (ICT) Journal*, 2018. https://doi.org/10.33260/zictjournal.v2i1.41

9. **Alinani Simukanga**, Madaliso Muhone, Joseph Mulenga, Jackson Phiri, & Mayumbo Nyirenda. Creation of an Agricultural Geographical Information System. *Zambia Information Communication Technology (ICT) Journal*, 2018. https://doi.org/10.33260/zictjournal.v2i2.57

Grants & Projects

- **Project SEARCH**, Researcher/Application Developer. Collaborative project with Boston University and University of Zambia (2018–2023). Research grant from the Fogarty International Center at NIH (R21TW010939).

Research Artefacts

1. **Alinani Simukanga** and Wenda Qin. SEARCH Software. 1, Zenodo, 11 Nov. 2021, https://doi.org/10.5281/zenodo.5676294.
2. **Alinani Simukanga** and Lauren Etter. Ear Datasets - Longitudinal. 1, Zenodo, 11 Nov. 2021, https://doi.org/10.5281/zenodo.5676103.
3. **Alinani Simukanga**, Lauren Etter, Misaki Kobayashi and Wenda Qin. Extended Data. 1, Zenodo, 11 Nov. 2021, https://doi.org/10.5281/zenodo.5675940.

Service & Administration

- **Departmental Leadership**: Acting Head of Department (Interim), University of Zambia.
- **Program Administration**: Former Program Coordinator for undergraduate and postgraduate programs.
- **Quality Assurance**: Contributed to course moderation, assessment quality checks, and curriculum implementation follow-through.
- **Student Development**: Coordinated internship-related support and supervised student project execution.

Awards & Honors

- **Best Poster Award**, IEEE International Joint Conference of Biometrics (IJCB) 2023
- **Jain Association Prize - Best Graduating Male Student in the School of Natural Sciences**, University of Zambia 2015
- **Dean's Prize - Best Overall Third-Year Student**, University of Zambia 2014

Professional Membership

- **Professional Engineer**, the Engineering Institution of Zambia (EIZ)
  EIZ No.: SMENG2025014377
- **Registered Engineer**, the Engineering Registration Board (EngRB)
  EngRB No.: SMENG2025014377
- **Full Member**, the ICT Association of Zambia (ICTAZ)
  Membership No.: FM003860

Certifications

- **Introduction to Cybersecurity**, Cisco Networking Academy 2022
- **Introduction to Modern AI**, Cisco Networking Academy 2026

Skills

Programming: `Python`, `C++`, `Java`.

Software & Systems: Software engineering workflows, systems analysis, and applied Android development.

Academic Practice: Curriculum delivery, project supervision, assessment moderation, and academic advising.

Research: Applied machine learning, biometrics research workflows, and publication development.

References

References can be provided on request.